Robust Remote Sensing Image Cross-Scene Classification Under Noisy Environment

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***Abstract*— In recent years, great progress has been made in the field of cross-scene classification. However, most existing cross-scene methods assume that the source domain has massive and carefully annotated data, which is time-consuming and labor-intensive in practice. Datasets in real-life applications usually contain a large number of noisy labels, which will significantly affect cross-scene classification performance. How to perform more discriminative and generalized cross-scene classification in the presence of noisy samples needs to be urgently addressed. Apart from that, existing methods tend to implement global matching between domains, causing problems such as unbalanced adaptation and negative transfer, limiting the cross-scene performance of the model. For more effective and reliable cross-scene classification under noisy environment, robust adaptation with noise (RAN) is proposed in this article. RAN explores which samples are noiseless and transferable to enable positive and robust cross-scene transfer. The curriculum learning strategy is used to filter out noisy samples for better source supervised learning and cross-domain matching. To further improve the stability and effectiveness of cross-scene adaptation, the class weighting factor and the public weighting factor are introduced to consider the class information of the source and target domains. RAN is an efficient plug-and-play adaptation framework, which is easily implemented and can be embedded in existing methods. Experimental results demonstrate that the proposed RAN can achieve remarkable performance on cross-scene classification tasks in noisy environments.**

***Index Terms*— Cross-scene classification, deep learning, domain adaptation (DA), noisy label.**

I. Introduction

IN RECENT years, scene classification has gained increasing attention in the field of remote sensing (RS). It attempts to build models that can discriminate the scene category of RS images. Not only does scene classification provide clues for other RS tasks, but also it is used in many real-world applications such as land use [1], urban planning [2], and environment and climate monitoring [3]. However, difficulty in acquiring high-quality labeled data and poor generalizability of models have seriously hampered the development and application of scene classification methods.

To address the above issues, RS cross-scene classification [4] is proposed to enhance the generalizability as well as the discriminative capability of the model to new data by leveraging existing knowledge. Domain adaptation (DA) is the most prevalent technique in cross-scene classification, which explores the relevance of different domains. Traditional (as opposed to deep learning) cross-scene classification methods use hand-crafted shallow features with poor representation capability, robustness, and generalizability. In contrast, deep neural networks (DNNs) can learn complex and deep features due to their powerful fitting capabilities [5]. Therefore, most of the latest cross-scene classification studies have been conducted based on DNNs [6].

Although DNNs have achieved impressive results in cross-scene tasks, their success relies on vast amounts of carefully labeled source data. In the field of RS, data labels can be extremely complex even for experienced RS experts. Thus, it is time-consuming and labor-intensive to acquire completely reliable source data in practice. Related surveys [7], [8], [9] show that noisy samples are very common in practical applications, and the ratio of corrupted labels can even be up to 30% in real-world datasets, as shown in Fig. 1. Zhang et al. [10] demonstrate that convolutional neural networks (CNNs) can easily fit with corrupted labels, which severely degrades the performance and generalization ability on test datasets. In addition, noisy labels will decrease the classifier's discriminative capability and increase the discrepancy between domain distributions, making the cross-scene transfer more difficult [11]. If noisy samples already exist in the source domain, existing methods will brutally align the corrupted samples with target data, leading to a significant reduction in model's performance.

Since most existing cross-scene approaches assume that the source domain is clean and noise-free, they are likely to fail in noisy environments. It is a key challenge to make the network learn robustly under noisy environments for RS cross-scene tasks. The critical of cross-scene classification with noisy labels can be divided into two steps. The first step is to distinguish between noisy and reliable samples in the source domain. Then, the reliable source samples and the target samples will participate in the adaptation process for cross-scene tasks. Except for not considering noisy samples, existing cross-scene methods still have some drawbacks. Most of them take a direct global alignment between the source and

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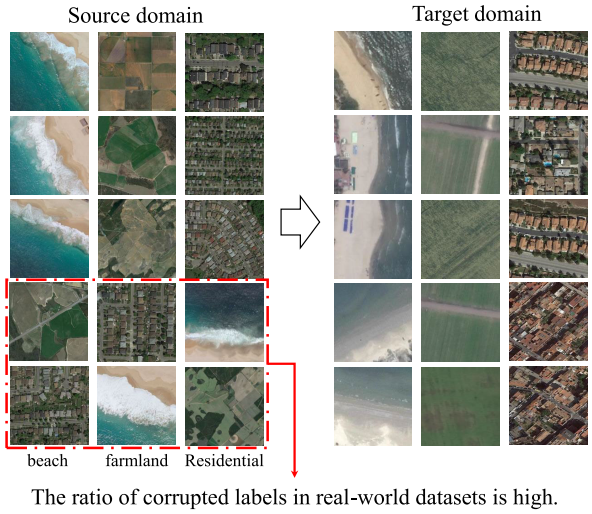


Fig. 1. Cross-scene classification tasks with noisy samples.

target domains, which is crude and blind with considerable potential risks. On the one hand, global alignment gives the same weight to different classes, and unbalanced adaptation can lead to a reduction in cross-scene performance. On the other hand, some categories appear in only one of the source or target domains during the training process. Blind alignment of all source and target samples, which may contain nonpublic categories, can explicitly cause the negative transfer.

To solve the above problems, this article presents a simple but effective method, robust adaptation with noise (RAN), for effective cross-scene classification under noisy environments. RAN tends to learn a model that eliminates the negative influence of noisy source samples and enables fine-grained cross-scene knowledge transfer at the class level. We first follow the curriculum learning strategy to filter out noisy samples and obtain reliable samples. Samples with smaller losses tend to be more reliable, while the opposite is more likely to be noisy. To further implement positive and fine-grained cross-scene adaptation, two weight factors are introduced by considering the class information for both the source and target domains. The class weighting factor reweights the samples according to the sample size of the different classes, to alleviate the unbalanced adaptation. The public weighting factor enables the detection of public classes to guarantee fine-grained adaptation at the class level. Then, the cross-scene model is extended with the selected reliable samples and the two weighting factors, to achieve a robust generalization from the noisy source domain to the unlabeled target domain.

RAN is an efficient plug-and-play cross-scene framework, which can be easily implemented and embedded with existing methods. To demonstrate the superiority of the proposed RAN, we conducted cross-scene classification tasks under different noisy environments based on four scene-classification datasets. In addition to quantitative experiments, we also do sufficient analysis for the proposed RAN. Experimental results show that the RAN outperforms existing methods by a large margin. The contributions of this article can be summarized as follows.

1) We propose adaptation framework RAN for more effective and robust cross-scene classification under noisy environments. Adopting the curriculum learning strategy and considering the class information via two weighting factors, RAN can improve the network's discriminative capability and generalizability of the network in cross-scene classification tasks under noisy environment.
2) RAN is an efficient plug-and-play framework and can be embedded with existing methods to improve their performance. Extensive experiments and analysis suggest that the proposed RAN outperforms existing methods by a large margin.

II. Related Works

Cross-scene classification in noisy environments is closely related to cross-scene classification, learning with noisy samples, and noisy DA. We thus provide a brief overview of the relevant areas.

A. Cross-Scene Classification

Cross-scene classification methods aim to use the knowledge learned from the source domain to improve the model's generalizability to the domain of interest (target domain), reduce the need for annotated data in the target domain, and lower computational resources for model training. With the rapid development of deep learning and the reveal of the transferability of DNNs [12], more and more cross-scene classification methods are using DNNs for feature extraction and cross-DA.

Mainstream methods can be divided into two main categories. First, discrepancy-based cross-scene methods [13], [14], [15], [16], [17] match the marginal or/and conditional distribution of extracted features across domains by adding adaptation layers in the network. The distribution metric is used to calculate the domain distribution discrepancy [e.g., maximum mean discrepancy (MMD)]. The trained network can explicitly reduce domain discrepancy and learn task-specific representations. Second, adversarial-based cross-scene methods [18], [19], [20], [21], [22] typically consist of a domain discriminator and a feature extractor. The former is trained to distinguish between source and target domain samples, and the latter is trained to confound the judgment of the domain discriminator. Therefore, the feature extractor can learn domain-invariant features.

Most of the existing cross-scene approaches assume that the source domains are clean without noise. But the fact is that noisy environments are common in the field of RS [23]. Existing cross-scene classification methods may be much less effective when faced with noisy samples.

B. Learning With Noisy Labels

Deep learning has made impressive achievements in many research and application areas, relying on vast amounts of high-quality data. However, it is difficult to guarantee the label quality of the data in real scenarios. Although many approaches and techniques have been proposed to deal with noisy samples, the overfitting problem of the network to noisy samples still exists and can impair the network's performance

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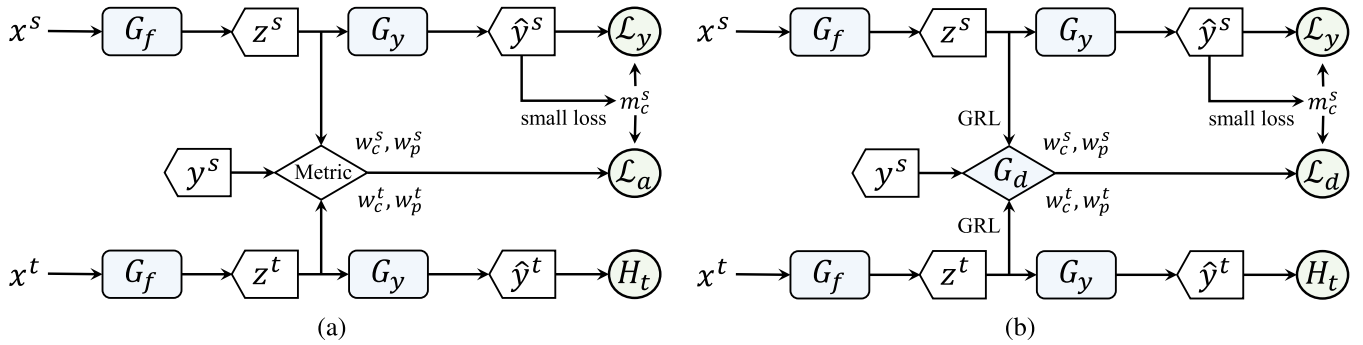


Fig. 2. Illustration of the proposed RAN. (a) RAN combined with metric-based approaches. (b) RAN combined with adversarial-based approaches. $\mathbf{z}$ is the feature representation extracted by feature extractor G_f. $\hat{y}$ is the class probability predicted by classifier G_y. GRL in (b) is the gradient reversal layer. The corresponding losses are the classification loss $\mathcal{L}_y$, adaptation loss $\mathcal{L}_a$ (or domain adversarial loss $\mathcal{L}_d$), and the target entropy loss H_t. The core idea of RAN is to use the noise mask m_c^s, the class weighting factor w_c, and the public weighting factor w_p to constrain the training of the network to improve its robustness to noisy samples as well as its discriminative capability and generalizability in cross-scene classification tasks.

and generalization ability [24]. Learning a robust and discriminative model with noisy datasets is therefore a very active area of research.

The relevant methods can be divided into four categories: 1) robust architecture-based approaches design specialized network architectures to accommodate different types of noisy labels, commonly by adding a noise adaptation layer to the network [25], [26]; 2) robust regularization-based approaches make the network less overfit to the noisy samples, either explicitly or implicitly, by designing regular terms [27], [28]; 3) robust loss-based approaches determine the confidence and weight of different samples in the loss calculation to reduce the impact of noisy labels on the network [29], [30]; and 4) sample selection-based approaches identify clean samples from the training data through multiple networks or multiple rounds of learning [31], [32].

C. Noisy DA

DA is concerned with matching and aligning the distribution of different domains. If noisy samples already exist in the source domain, existing methods would brutally align the wrong samples with the target domain, leading to a significant reduction in adaptation performance. Therefore, the critical of noisy DA is to distinguish the noisy samples and then use those reliable samples for cross-DA. Transferable curriculum learning (TCL) [33] is proposed to find the source samples which are noiseless and transferable. Han et al. [34] use offline curriculum learning to minimize a newly empirical source risk and propose a proxy distribution-based margin discrepancy to reduce the impact of feature noise. Zuo et al. [35] propose a model-agnostic module, called seek common ground while reserving differences (SCGWRDs), to deal with noisy samples by utilizing the outputs of two symmetrical DA models. Zhuo et al. [11] perform reliable sample selection by modeling uncertainty in the predicted probability of the classifier and reweighting the classification loss by this uncertainty.

III. Methods

The notations and problem settings are introduced first for describing the proposed method better. The source domain $\mathcal{D}_s = \{\mathbf{x}_i^s, y_i^s\}_{i=1}^{n_s}$ with n_s labeled samples may have corrupted labels. And the target domain $\mathcal{D}_t = \{\mathbf{x}_i^t\}_{i=1}^{n_t}$ is with n_t unlabeled samples. $\mathcal{D}_s$ and $\mathcal{D}_t$ are sampled from different data distributions p and q.

To implement robust and effective cross-scene classification tasks, we propose RAN, which is shown in Fig. 2. The framework is made up of the feature extractor G_f and the classifier G_y (and the domain discriminator G_d if the adversarial-based methods are used). $\hat{y}$ denotes the class probabilities predicted by G_y. And $\hat{y}^t$ denotes the target pseudolabels. Our goal is to train a DNN to eliminate the negative influence of noisy source samples and enable the positive and robust transfer between reliable source samples and target samples. To mitigate the influence of noise samples, we follow the curriculum learning strategy. For positive and robust transfer, we propose the class weighting factor and the public weighting factor. RAN is very lightweight and plug-and-play, which significantly improves the model's ability to generalize to the target domain under noisy environments.

A. Deep Cross-Scene Classification

As mentioned earlier, current cross-scene classification methods are mainly divided into metric-based and adversarial-based. The following is a brief description of the two types of methods.

Metric-based methods are an important part of the cross-scene classification. They mainly use adaptation layers with specific metrics and train the network jointly with the adaptation loss to learn domain-invariant features. The loss function for such methods can be summarized as

$$\min_{G_f, G_y} \mathcal{L}_y + \lambda \mathcal{L}_a \tag{1}$$

where $\mathcal{L}_y$ and $\mathcal{L}_a$ denote the classification loss and the adaptation loss, respectively. $\mathcal{L}_a$ is usually calculated via some kind of distribution metric, like MMD [36] and correlation distance (CORAL) [37].

Adversarial learning has also been successfully explored to minimize the distribution discrepancy. The domain discriminator G_d is trained to distinguish different domains, while feature extractor G_f tries to confuse the former. The loss function can be formalized as

$$\min_{G_f, G_y, G_d} \mathcal{L}_y + \lambda \mathcal{L}_d \tag{2}$$

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where $\mathcal{L}_y$ and $\mathcal{L}_d$ denote the classification loss and the domain adversarial loss. The initial training objective of adversarial-based methods is a two-player optimization procedure. And (2) can be obtained by introducing the gradient reversal layer (GRL) as a "pseudofunction" [38].

Existing deep cross-scene classification methods have the following drawbacks: 1) noisy samples will be involved in the domain alignment, and it will severely degrade the model's performance and generalization ability; 2) global alignment gives the same weight to different classes, and unbalanced classes can lead to a reduction in adaptation performance; and 3) direct alignment of all source and target samples in each mini-batch will lead to the negative transfer. Therefore, we propose RAN to alleviate the above problems.

B. Robust Adaptation With Noise

To distinguish the noisy samples, we follow the curriculum learning strategy [39] and generate a noise mask m_c^s for source samples by a loss threshold γ:

$$m_c^s(\mathbf{x}_i) = \mathbf{1}(l_i \leqslant \gamma), \quad i = 1, 2, \ldots, n_s \tag{3}$$

where $\mathbf{1}$ is the indicator function and l_i is the classification loss of $\mathbf{x}_i^s$. When the sample's loss is smaller than the threshold γ, it will be selected as reliable sample for training. Conversely, if the sample has a large loss, we consider it to be a noisy sample. The classifier is then only trained with these reliable samples to improve the model's discriminative capability. In addition, the adaptation process will be carried out in reliable source samples and target samples.

Some studies select reliable samples by a certain rate [34]. However, the ratio of corrupted labels in real-world datasets is reported to range from 8.0% to 38.5% [24]. A fixed ratio may result in noisy samples not being filtered completely, or some reliable samples being filtered. Therefore, we select reliable samples by a threshold in this article for more stable training.

The class weighting factor w_c is introduced to reweight the samples according to the sample size of the corresponding class, which is different from global alignment. The larger the sample size of the class to which the sample belongs, the smaller the weight will be assigned. Conversely, the sample will have a higher weight, if it belongs to a class with a smaller sample size. The domination of the adaptation process by categories with large sample sizes can be suppressed, thus mitigating the problem of imbalanced sample adaptation. For the source domain, it is easy to calculate $w_c^s(\mathbf{x}_i^s) = y_{ic}^{\text{one-hot}} / \sum_{(\mathbf{x}_j, y_j)\in\mathcal{D}_s} y_{jc}^{\text{one-hot}}$ by converting the source labels to one-hot vectors, where y_{ic} is the cth term in $\mathbf{y}_i^{\text{one-hot}}$, corresponding to the cth class. For the target domain, the pseudolabels obtained from the model predictions can be used in the same way to calculate $w_c^t(\mathbf{x}_i^t) = \hat{y}_{ic}^{\text{one-hot}} / \sum_{(\mathbf{x}_j, \hat{y}_j)\in\mathcal{D}_t} \hat{y}_{jc}^{\text{one-hot}}$ in any case. However, wrong hard predictions may affect the model's performance. The class prediction probability is a better choice to take the uncertainty into consideration $w_c^t(\mathbf{x}_i^t) = \hat{y}_{ic} / \sum_{(\mathbf{x}_j, \hat{y}_j)\in\mathcal{D}_t} \hat{y}_{jc}$.

The public weighting factor w_p is introduced to guarantee that the adaptation process is implemented during samples in the public class. When a class does not appear in both the source and target domains, we will not blindly allow such samples to participate in the adaptation process, which leads to negative transfer. It makes the adaptation more fine-grained at class-level, which is different from that coarse-grained global alignment. Let us set the common appeared classes as $\text{class}_p = \text{set}(y^s) \cap \text{set}(y^t)$. The source weight can be directly calculated by the use of source labels $w_p^s(\mathbf{x}_i) = \mathbf{1}(y_i^s \in \text{class}_p)$. And the target pseudolabel is used for calculating the target weight $w_p^t(\mathbf{x}_i) = \mathbf{1}(\hat{y}_i^t \in \text{class}_p)$. It is worth noting that this weighting factor is more effective for metric-based methods. This is because these methods calculate between each sample to obtain the distribution discrepancy. For other, e.g., adversarial-based methods, this weighting factor has some boosting effect, but not significant enough.

Combining the mask m_c^s and weighting factors w^s and w^t, the network can eliminate the negative impact of noisy labels and enable robust and positive cross-scene knowledge learning. The overall weights of the source and target domains can be formulated as follows:

$$\begin{aligned} w^s &= w_c^s * w_p^s * m_c^s \\ w^t &= w_c^t * w_p^t. \end{aligned} \tag{4}$$

C. Implementation Details

By filtering out the noisy samples, the classifier is trained with reliable source samples. The discriminative capability of the network can be guaranteed in this case. The extended classification loss can be formulated as

$$\mathcal{L}_y = \frac{1}{n_s} \sum_{\mathbf{x}_i\in\mathcal{D}_s} m_c^s(\mathbf{x}_i) L_{\text{ce}}(\hat{y}_i, y_i) \tag{5}$$

where L_{ce} is the cross entropy loss and $m_c(\mathbf{x}_i)$ indicates whether the sample is reliable or not.

Next, we show the implementation details of the proposed method using the standard metric-based approach and adversarial-based approach as examples, which are also shown in Fig. 2.

1) Metric-Based: We choose MMD [36] as an example of this type of method. MMD is a nonparametric distance estimate between two distributions, which has been widely used in the field of cross-scene classification. It can be formulated as follows:

$$\hat{d}(p, q) = \left\| \frac{1}{n_s} \sum_{\mathbf{x}_i\in\mathcal{D}_s} \varphi(\mathbf{x}_i) - \frac{1}{n_t} \sum_{\mathbf{x}_j\in\mathcal{D}_t} \varphi(\mathbf{x}_j) \right\|_{\mathcal{H}}^2 \tag{6}$$

where φ is a nonlinear mapping function.

MMD is a coarse-grained distribution metric, calculated with equal weights between each source and target sample. This can lead to unbalanced adaptation and negative transfer. Combined with the proposed RAN, MMD can be more effective and robust at fine-grained level. The new adaptation loss $\mathcal{L}_a$ is equal to the extended MMD

$$\mathcal{L}_a = \left\| \frac{1}{n_s} \sum_{\mathbf{x}_i\in\mathcal{D}_s} w_i^s \varphi(\mathbf{x}_i) - \frac{1}{n_t} \sum_{\mathbf{x}_j\in\mathcal{D}_t} w_j^t \varphi(\mathbf{x}_j) \right\|_{\mathcal{H}}^2 \tag{7}$$

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Fig. 3. Samples from datasets. (a) AID. (b) CLRS. (c) MLRSN. (d) OPTIMAL-31. From left to right, they are *airport, beach, bridge, commercial, desert, farmland, forest, industrial, meadow, mountain, overpass, parking, playground, port*, and *residential*.

where w_i^s and w_j^t denote the source weight and target weight for samples $\mathbf{x}_i$ and $\mathbf{x}_j$.

Entropy can be treated as a measurement of the prediction uncertainty. Thus, we use the entropy minimization principle to quantify the target prediction, which is also a guarantee of the target prediction probabilities and pseudolabels. And it can be calculated as

$$H_t = -\sum_{\hat{y}_i \in \mathcal{D}_t} \hat{y}_i \log \hat{y}_i. \tag{8}$$

The final objective function of metric-based approaches is made up of the classification loss $\mathcal{L}_y$, the adaptation loss $\mathcal{L}_a$, and the entropy loss H_t, and it can be formulated as

$$\min_{G_f, G_y} \mathcal{L}_y + \lambda \mathcal{L}_a + \beta H_t \tag{9}$$

where λ is the adaptation factor and β is the tradeoff parameter.

2) Adversarial-Based: We choose domain adversarial neural network (DANN) [38] as an example of this type of method. The standard domain adversarial loss can be formulated as

$$\mathcal{L}_d = \frac{1}{n_s + n_t} \sum_{\mathbf{x}_i \in (\mathcal{D}_s \cup \mathcal{D}_t)} L_{\text{bce}}(G_d(G_f(\mathbf{x}_i)), d_i) \tag{10}$$

where L_{bce} is the binary cross entropy loss. d_i means the domain label of the input samples (1 and 0 for the source and target domains, respectively).

The adversarial training process described above is straightforward and rough as well. And in combination with the proposed RAN, the adaptation can be more stable and positive. Since the source and target samples will be classified into two domains by the domain discriminator, w_p will not be as effective as in the metric-based approach described above. The new domain adversarial loss can be extended as

$$\mathcal{L}_d = \frac{1}{n_s} \sum_{\mathbf{x}_i \in \mathcal{D}_s} w_i^s L_{\text{bce}}(G_d(G_f(\mathbf{x}_i)), d_i) + \frac{1}{n_t} \sum_{\mathbf{x}_j \in \mathcal{D}_t} w_j^t L_{\text{bce}}(G_d(G_f(\mathbf{x}_j)), d_j). \tag{11}$$

It is worth noting that the adversarial training process could be efficient with the GRL. The final objective function of adversarial-based approaches is made up of the classification loss $\mathcal{L}_y$, the domain adversarial loss $\mathcal{L}_d$, and the entropy loss H_t, and it can be formulated as

$$\min_{G_f, G_y, G_d} \mathcal{L}_y + \lambda \mathcal{L}_d + \beta H_t. \tag{12}$$

RAN is an effective cross-scene classification framework, which is easy to implement. By minimizing (10) or (12), we can improve the model's discriminability and generalizability by harnessing noisy samples and mitigating negative transfer at the class level. As DNNs fit clean samples before fitting noisy samples [40], the noisy mask m_c will not be used in the initial stages of training, and the same goes for other methods involving noise processing.

IV. Experiments

We evaluate RAN with some state-of-the-art methods on cross-scene classification tasks under noisy environments. Experimental results and analysis demonstrate that the proposed RAN outperforms existing methods by a large margin.

A. Datasets and Experimental Setup

1) Cross-Scene Datasets: Four scene-classification datasets are used in our experiments, including AID [41], CLRS [42], MLRSN [43], and OPTIMAL-31 [44], which is also shown in Fig. 3. Fifteen shared categories of images are selected from each dataset above. The categories are *airport, beach, bridge, commercial, desert, farmland, forest, industrial, meadow, mountain, overpass, parking, playground, port*, and *residential*. In particular, not all samples were selected in our datasets. For domains with particularly large sample sizes, we select 100–200 images per class to make the cross-scene classification tasks more challenging. AID, CLRS, MLRSN, and OPTIMAL-31 datasets are set as domains A, C, M, and O, respectively. In each task, we choose two of the four domains as the source domain and the target domain (like A-C). There are therefore a total of 12 cross-scene classification tasks in each setting.

2) Compared Methods: The proposed RAN is compared with the following state-of-the-art methods: 1) baseline: ResNet [45]; 2) DA methods: deep adaptation networks (DANs) [36], DANNs [38], deep correlation (CORAL) [37], margin disparity discrepancy (MDD) [46], semantic concentration for DA (SCDA) [47], and discriminator-free adversarial

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TABLE I
CLASSIFICATION ACCURACY (%) WITH 45% SYMMETRIC NOISY LABEL

Method	A-C	A-M	A-O	C-A	C-M	C-O	M-A	M-C	M-O	O-A	O-C	O-M	avg
ResNet	64.56	61.46	73.12	67.74	51.63	64.90	61.75	47.27	64.16	57.69	47.44	50.40	59.34
CORAL	50.39	45.89	49.58	61.79	47.61	55.21	50.35	61.79	53.54	57.47	44.89	48.04	52.21
SCDA	50.22	50.35	60.10	79.59	42.14	66.04	71.62	46.28	69.27	76.64	51.50	61.40	60.43
DALN	70.89	69.19	66.25	90.80	72.57	72.50	76.42	52.61	70.10	89.05	59.39	71.82	71.80
SPL	69.33	63.39	72.29	44.25	34.98	49.27	47.37	40.78	51.77	57.39	47.50	48.85	52.26
Mentor	47.33	44.61	47.08	59.96	45.73	58.12	48.94	39.27	53.33	57.09	47.22	48.79	49.79
TCL	69.72	76.70	71.14	91.73	69.99	80.41	88.23	55.16	75.10	84.32	61.55	69.08	74.63
RDA	73.11	70.04	82.08	93.85	62.90	87.50	78.06	60.06	71.48	87.48	53.83	74.55	74.58
DAN	53.00	53.24	58.64	70.68	50.83	61.56	56.91	44.39	57.50	68.82	49.50	61.19	57.19
RAN+MMD	**83.06**	**83.09**	86.56	**95.87**	77.94	88.44	**95.42**	**63.89**	82.08	**95.27**	**69.56**	**82.45**	**83.64**
DANN	54.50	56.09	56.67	77.65	61.30	66.04	70.13	43.89	61.14	76.16	52.78	61.19	61.46
RAN+DANN	79.44	82.98	88.33	95.34	**78.90**	89.58	90.24	59.05	81.88	89.05	68.28	79.44	81.88
MDD	57.50	67.30	66.35	86.74	73.75	87.71	82.12	43.94	67.39	79.21	48.61	63.39	68.67
RAN+MDD	82.17	76.44	**92.60**	95.49	72.03	**92.39**	86.96	61.72	**88.64**	94.30	65.89	77.08	82.14

TABLE II
CLASSIFICATION ACCURACY (%) WITH 20% SYMMETRIC NOISY LABEL

Method	A-C	A-M	A-O	C-A	C-M	C-O	M-A	M-C	M-O	O-A	O-C	O-M	avg
ResNet	77.78	72.46	81.46	89.09	66.99	78.96	76.57	60.22	77.29	83.13	67.50	74.40	75.49
CORAL	63.27	65.38	72.92	81.68	63.61	76.46	70.73	55.05	72.29	78.29	64.83	66.62	69.26
SCDA	59.67	69.24	85.52	86.33	70.75	77.50	74.97	63.61	78.54	89.50	71.28	68.92	74.65
DALN	79.39	74.77	82.50	93.71	80.84	88.23	82.83	65.72	76.98	92.66	73.01	83.20	81.15
SPL	79.00	74.93	83.23	85.44	64.20	81.88	72.40	60.00	76.67	82.16	65.56	68.38	74.49
Mentor	62.11	63.55	74.06	81.94	64.20	77.60	67.45	55.11	71.67	78.36	66.61	67.95	69.22
TCL	80.27	81.15	86.45	96.20	78.58	90.62	84.28	63.00	82.18	91.39	76.11	81.21	82.62
RDA	79.65	76.65	90.79	95.38	80.79	91.04	80.52	62.83	82.08	90.50	72.11	79.17	81.79
DAN	67.78	71.01	78.54	85.88	69.51	80.00	76.95	58.94	75.63	85.88	70.72	74.13	74.58
RAN+MMD	**86.83**	**84.86**	92.29	96.65	**82.55**	92.71	**96.20**	70.94	**86.04**	96.69	**83.22**	**86.47**	**87.95**
DANN	71.44	67.79	78.33	88.98	72.14	81.46	76.46	58.78	78.75	86.93	70.33	78.37	75.81
RAN+DANN	85.56	83.72	90.10	**97.73**	82.34	**93.44**	93.15	66.17	84.96	**96.83**	81.11	84.16	86.61
MDD	66.56	72.41	77.71	92.21	73.59	88.33	77.59	56.33	73.33	85.88	68.44	74.82	75.60
RAN+MDD	85.56	81.64	**94.27**	96.87	82.02	93.02	94.71	**71.11**	85.00	94.00	82.89	84.28	87.11

TABLE III
CLASSIFICATION ACCURACY (%) WITH 45% PAIRFLIP NOISY LABEL

Method	A-C	A-M	A-O	C-A	C-M	C-O	M-A	M-C	M-O	O-A	O-C	O-M	avg
ResNet	41.00	39.39	43.22	47.48	41.65	44.79	36.42	34.50	41.35	40.22	35.05	40.15	40.44
CORAL	40.22	34.73	41.15	50.86	43.21	44.27	43.65	36.56	44.48	42.46	35.72	40.58	41.49
SCDA	40.56	36.55	41.46	55.20	48.96	50.89	49.72	32.72	45.10	59.59	39.06	47.66	45.62
DALN	42.11	41.22	44.90	73.33	51.21	45.21	49.16	39.72	45.00	47.45	49.44	56.41	48.76
SPL	42.56	49.38	39.17	47.56	39.02	42.40	40.33	35.27	41.77	45.47	37.94	43.42	42.02
Mentor	41.23	37.14	41.67	48.49	44.44	46.25	44.88	35.61	43.85	42.79	39.61	45.14	42.59
TCL	44.66	47.82	50.93	68.23	52.17	53.22	53.81	45.27	53.85	58.36	46.44	52.76	52.29
RDA	46.83	47.87	45.52	62.68	51.17	46.77	53.29	41.45	57.50	52.88	47.50	54.21	50.64
DAN	40.94	39.99	43.44	52.96	46.43	47.08	45.10	36.06	46.67	46.70	39.39	46.45	44.27
RAN+MMD	**54.00**	**59.26**	62.81	73.93	**61.89**	**63.13**	60.74	47.39	63.75	**59.96**	52.50	**57.76**	59.76
DANN	43.88	42.03	44.69	56.68	48.74	46.88	42.35	39.61	54.06	51.06	43.56	52.22	47.15
RAN+DANN	51.22	57.11	60.31	**76.91**	54.91	60.73	65.88	**51.44**	60.93	59.92	51.06	54.69	58.76
MDD	46.27	53.99	61.97	60.81	42.56	55.20	49.12	35.17	46.96	47.03	35.33	44.17	48.22
RAN+MDD	51.94	55.45	**72.39**	75.45	57.05	62.60	**72.70**	48.11	**66.15**	58.66	**54.22**	55.02	**60.81**

learning network (DALN) [48]; 3) noise-based methods: self-paced learning (SPL) [39] and MentorNet [31]; and 4) noise-based DA methods: TCL [33] and robust DA (RDA) [34]. In particular, the proposed RAN is combined with DAN, DANN, and MDD as RAN + MMD, RAN + DANN, and RAN + MDD, respectively.

3) Label Corruption: Since the above datasets are only with a small amount of noise, we manually corrupted the data labels to simulate different noise environments. We are mainly concerned with label corruption in this article. A typical approach for modeling label noise assumes that the corruption process is conditionally independent of data features when the true label is given [24]. The true label is corrupted by a noise transition matrix, which denotes the probability of the true label being flipped into the corrupted label. Based on the label corruption, we create noisy source domains from the original clean dataset in two different ways: symmetric noise and pairflip noise. For symmetric noise, a true label is flipped into other labels with equal probability. For pairflip noise, a true label is flipped into only a certain label. As an example, if the source domain is with 45% symmetric noise, it represents that 45% of the source labels are corrupted by symmetric noise.

4) Experimental Settings: We follow the standard evaluation protocols for cross-scene classification and use all labeled source samples and unlabeled target samples for training. All the models and results are implemented based on the Pytorch framework. The pretrained ResNet-50 [45] is used as the base model for all methods, and a fully connected

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TABLE IV
CLASSIFICATION ACCURACY (%) WITH 20% PAIRFLIP NOISY LABEL

Method	A-C	A-M	A-O	C-A	C-M	C-O	M-A	M-C	M-O	O-A	O-C	O-M	avg
ResNet	70.33	68.65	78.43	80.29	63.98	76.67	69.12	54.06	73.75	71.83	59.78	64.41	69.28
CORAL	64.83	63.18	68.02	76.50	62.90	72.92	66.29	53.17	71.56	69.23	57.11	64.30	65.83
SCDA	61.78	62.32	80.83	89.42	71.12	76.35	67.44	48.33	72.60	87.00	62.11	73.59	71.07
DALN	76.94	82.13	81.56	91.69	77.13	84.27	82.64	64.83	71.17	88.79	76.39	79.01	79.71
SPL	74.00	74.07	80.10	72.06	60.11	77.29	63.98	53.39	71.04	72.10	58.89	63.82	68.40
Mentor	62.77	64.03	67.39	73.52	61.72	73.54	67.11	52.61	68.54	67.60	58.61	65.59	65.25
TCL	75.27	76.43	83.22	94.63	71.39	87.18	83.50	63.72	82.70	85.54	74.94	78.13	79.72
RDA	74.83	78.85	81.25	89.12	78.47	88.23	80.37	59.05	79.16	90.09	73.66	78.42	79.29
DAN	66.39	71.60	77.81	81.22	66.93	77.50	75.56	55.94	78.29	79.18	63.33	71.82	72.13
RAN+MMD	83.17	**84.92**	**93.02**	96.54	**80.35**	92.81	82.98	66.39	85.42	**94.53**	**79.89**	**84.49**	85.38
DANN	71.33	74.02	75.83	83.24	74.07	79.38	75.01	61.39	80.83	80.11	70.50	76.49	75.18
RAN+DANN	81.83	84.11	91.35	**97.83**	79.28	**93.54**	**85.32**	67.11	82.71	93.37	78.22	80.21	84.57
MDD	66.44	70.75	72.60	86.36	66.82	85.41	70.42	52.00	72.39	78.32	57.38	76.65	71.30
RAN+MDD	**83.28**	84.54	91.35	96.53	80.19	93.23	84.58	**69.78**	**85.83**	93.37	79.78	83.31	**85.48**

TABLE V
ABLATION STUDY (%) WITH 45% SYMMETRIC NOISY LABEL

Method	A-C	A-M	A-O	C-A	C-M	C-O	M-A	M-C	M-O	O-A	O-C	O-M	avg
RAN w/o H_t	79.11	78.31	83.02	93.41	76.38	82.39	82.05	60.33	75.31	93.25	66.38	72.51	78.54
RAN w/o m_c	78.22	79.87	79.68	92.96	75.89	80.42	85.02	60.27	71.25	89.76	67.17	78.52	78.25
RAN w/o $w_c + w_p$	79.11	79.22	**90.31**	92.96	74.39	83.13	89.19	60.78	77.08	89.23	63.44	77.24	79.67
RAN w/o w_c	80.44	80.30	86.14	94.07	76.70	86.25	88.90	61.27	79.17	92.92	68.66	78.20	81.09
RAN w/o w_p	82.72	78.42	86.04	95.38	77.51	85.42	95.19	61.12	81.67	93.70	68.16	80.84	82.18
RAN	**83.06**	**83.09**	86.56	**95.87**	**77.94**	**88.44**	**95.42**	**63.89**	**82.08**	**95.27**	**69.56**	**82.45**	**83.64**

bottleneck layer is added before its classifier layer. Since the feature extractor G_f is pretrained, the learning rate is set to one-tenth of the classifiers G_y and other modules. For a fair comparison, the training setups for the different methods are basically the same, while the specific parameters for their proposed loss or regularization items are selected according to the best values in their articles. We use mini-batch stochastic gradient descent (SGD) with batch size 32, momentum 0.9, and weight decay 0.0005. The model is trained with an initial learning rate of 0.01, and the learning rate also changes with an annealing strategy [38]. The image's size will be adjusted to the corresponding input size of each method when training and inferring. For MMD-based methods, we adopt Gaussian kernel with bandwidth set to median pairwise squared distances on the training data.

B. Results

The quantitative evaluation criterion is the accuracy of the model to classify the target sample. The results of the cross-scene classification tasks under 45% symmetric noisy label, 20% symmetric noisy label, 45% pairflip noisy label, and 20% pairflip noisy label are shown in Tables I–IV, respectively. It can be seen that RAN outperforms all the comparison methods by a large margin on all the tasks. Compared to DA methods TCL and RDA, which also consider noisy samples, RAN + MMD still has certain advantages. In addition, RAN + DANN and RAN + MDD are also very competitive, outperforming the comparison methods on most tasks and even achieving the best performance on some tasks.

More specifically, it is difficult for ResNet to learn from the noisy source domain and generalize well to the target domain. However, it is widely used as a benchmark to show the strengths and weaknesses of other methods. Commonly used adaptation methods, such as DAN, DANN, CORAL, MDD, and DALN, suffer from negative transfer and overfitting caused by noisy source samples, which will reduce the discriminative capability of the classifier and disrupt the adaptation process. That is why they perform even worse than the baseline ResNet on some tasks. The reason why SCDA, a method that performs very well in conventional settings, is also very limited in noisy environments is because SCDA requires the construction of intraclass sample pairs between the source and target domains. SPL and MentorNet can learn more robustly from noisy samples but did not make connections between domains in cross-scene tasks. TCL and RDA not only try to filter noisy samples but also bridge the source and target domains. However, their crude global cross-domain matching also makes the result not ideal enough. The proposed RAN not only mitigates the affection of source noisy samples but also performs more reliable and positive cross-scene knowledge transfer at the class level. Therefore, there is a huge improvement in all tasks compared to other methods.

By comparing the results in the symmetric noise environment with those in the pairflip noise environment, it can be seen that the latter is more challenging than the former. This is because the symmetric noise is spread over multiple classes so that each class is hardly fatal to the correct one. In contrast, pairflip noise samples are all in a certain class, so it can have a greater impact on the discriminative capability of the model.

C. Analysis

In this section, we use the RAN + MMD as the main example for relevant experimental analysis and discussion.

1) Ablation Study: We investigate different modules of RAN by ablation study: 1) RAN w/o H_t removes the target entropy loss from the optimization function; 2) RAN w/o m_c means that the noise mask m_c does not participate in the

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TABLE VI
CLASSIFICATION ACCURACY (%) WITH CLEAN LABEL

Method	A-C	A-M	A-O	C-A	C-M	C-O	M-A	M-C	M-O	O-A	O-C	O-M	avg
ResNet	78.59	75.46	84.69	92.29	71.66	85.10	80.14	63.56	81.04	85.70	70.78	78.15	78.93
CORAL	80.39	76.60	84.90	91.96	72.62	85.10	80.48	66.06	82.50	87.37	73.11	78.31	79.95
SCDA	84.22	82.39	87.71	97.13	81.16	88.85	91.05	**78.12**	87.71	92.74	84.28	86.97	86.86
DALN	87.44	86.15	89.17	96.08	**85.08**	90.73	81.75	73.36	80.73	95.75	83.61	87.87	86.48
SPL	82.67	77.78	84.89	93.48	73.53	88.12	77.09	65.89	79.47	87.03	72.39	76.92	79.94
Mentor	78.77	77.13	85.52	93.56	73.91	86.67	79.92	65.16	79.48	88.16	73.50	79.60	80.12
TCL	86.55	84.11	87.92	97.28	79.06	91.17	86.63	67.94	84.48	93.18	81.11	83.73	85.26
RDA	81.72	82.15	87.60	96.72	78.53	93.48	83.65	65.94	92.39	92.17	73.34	81.11	84.07
DAN	83.22	81.80	88.75	93.10	78.58	88.02	87.23	72.22	89.79	91.58	79.72	81.21	84.60
RAN+MMD	87.83	**86.90**	92.81	97.65	84.32	95.00	85.69	76.28	88.12	**97.39**	**85.89**	87.60	**88.79**
DANN	85.78	78.21	87.60	94.00	83.79	87.19	88.12	69.38	**92.50**	93.15	81.38	84.92	85.50
RAN+DANN	87.17	84.60	90.42	**98.03**	83.36	94.58	91.73	71.78	84.38	96.01	85.22	**93.89**	88.43
MDD	82.94	82.50	89.16	95.38	81.26	93.33	90.13	72.44	86.56	75.75	69.27	82.93	83.47
RAN+MDD	**88.67**	84.86	**95.42**	97.21	82.29	**95.42**	**93.33**	73.94	87.71	94.39	85.77	84.97	88.67

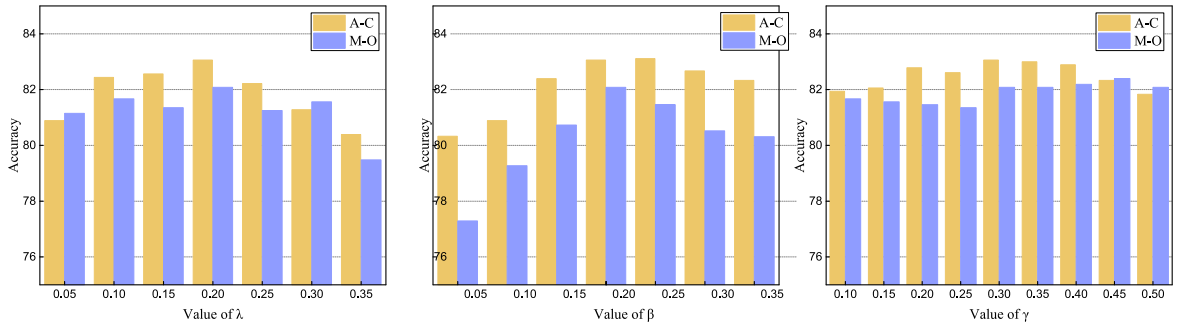


Fig. 4. Parameter sensitivity analysis on tasks A-C and M-O with 45% symmetric noise. (a) Sensitivity of λ. (b) Sensitivity of β. (c) Sensitivity of γ.

adaptation process (also contained in the classification loss); 3) RAN w/o w removes the two weighting factors from the adaptation process; 4) RAN w/o w_c removes the class weighting factor from the adaptation process; and 5) RAN w/o w_p removes the public weighting factor from the adaptation process. The results of our ablation study are shown in Table V. It quantifies the performance of the various components of the RAN and demonstrates that how to learn robustly with both noise and fine-grained adaptation is very important for cross-scene classification tasks.

2) Clean Environment: To verify the superiority of RAN, we conducted cross-scene experiments with clean source domains (in line with the usual cross-scene classification task setup), shown in Table VI. Although RAN is proposed to deal with cross-scene classification tasks in noisy environments, it can still achieve a huge advantage in the standard noiseless cross-scene classification task. In contrast, TCL and RDA have no obvious advantage over standard adaptation methods such as DAN and DANN. This demonstrates the versatility of RAN.

3) Parameter Sensitivity Analysis: We evaluate the sensitivity of hyperparameters on tasks A-C and M-O with 45% symmetric noise, shown in Fig. 4.

Results of λ and β are shown in Fig. 4(a) and (b). The best performance of RAN + MMD is obtained with a value of around 0.2 for both parameters in this setting. In fact, the optimum value of the hyperparameter varies with the noise level in practice. It is difficult for us to find the same value to achieve the best performance for all settings and tasks. In this experiment, we set both parameters to the value of 0.2 for all settings.

Since γ is important in separating the noisy and reliable samples, we further explore its sensitivity in Fig. 4(c). The impact of γ on RAN varies with the task as well. The reason is that the difficulty of supervision for different source domains and the difficulty of adaptation for different target domains are not the same. Besides, the RAN filters out the noisy sample by its classification loss, and the ability of noisy samples to affect different tasks is also different. However, RAN is relatively robust to this threshold γ within a certain range. Considering its performance on all tasks, we finally set it to 0.3 for all settings.

4) Convergence: The training error and test accuracy of different methods on task C-A with 45% symmetric noise are shown in Fig. 5(a) and (b). Note that the training error and test accuracy are recorded for every 10 and 500 epochs, respectively. Although most methods, with the exception of ResNet, MDD, and RDA, converge quickly on source domain samples, their performance on the target domain is not as good. In addition to the bad effect of noisy samples, the lack of robustness of their adaptation process is also an important factor. RAN can handle noisy samples while being more robust and effective at cross-scene classification tasks. As a result, RAN-based approaches not only converge quickly on the source domain but also generalize quickly to the target domain. RAN is not only fast in convergence but also very stable in training.

There are also some insightful findings.

1) RDA and MDD, which are both based on the MDD metric, cannot converge well on the source domain. As you can see from the test accuracy curves, they are also very

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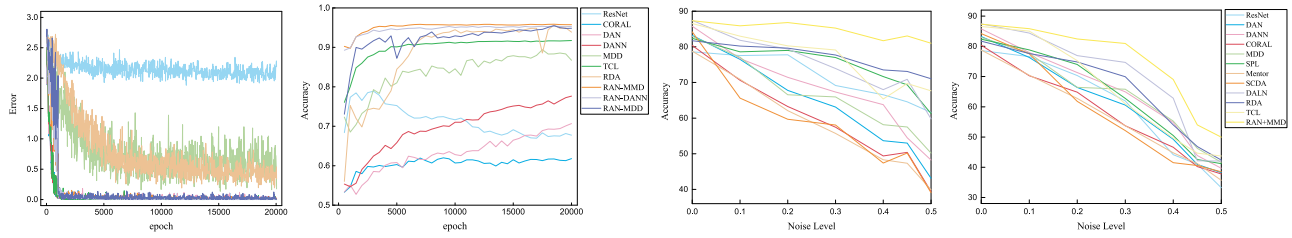


Fig. 5. (a) and (b) Training error curves and test accuracy curves on task C-A with 45% symmetric noise. (c) and (d) Classification accuracy with respect to different noise levels on task A-C.

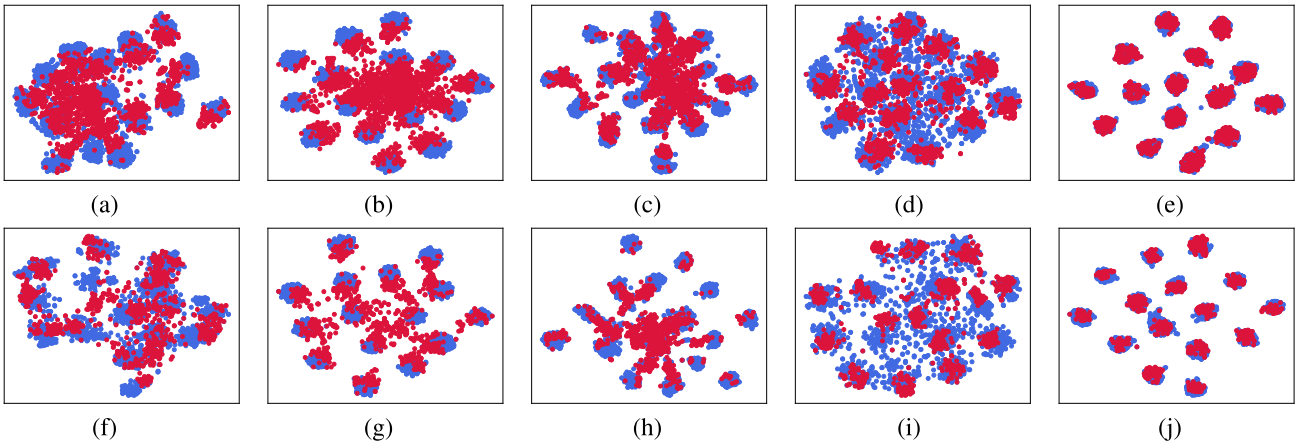


Fig. 6. Visualizations of the learned features using t-SNE embeddings with 45% symmetric label. (a)–(e) are from Task A-C. (f)–(j) are from Task M-O. The blue and red parts denote the source and target samples.

instable when training. However, RAN-MDD has a better performance, which demonstrates the effectiveness of RAN.

2) It can be seen that the test accuracy of DAN and DANN is still increasing. In fact, they will probably reach convergence at 40k–50k epochs, and the accuracy will be about 2% higher than now. In contrast, RAN + MMD and RAN + DANN (corresponding to the DAN and DANN) reach convergence in less than 10k epochs with impressive performance.
3) Let us finally focus on ResNet's test accuracy curve. The accuracy first rises briefly during the training process and then falls gradually, which is consistent with the previous statement in Section III. As DNNs fit clean samples before fitting noisy samples [40], we choose to have all source domain samples participate in the training at the beginning and deal with the noisy samples later (1000 epochs in our experiments).

5) Noisy Level: Fig. 5(c) and (d) records the results in various noise levels on A-C tasks (both symmetric and pairflip noise). Noise level contains {0, 0.1, 0.2, 0.3, 0.4, 0.45, 0.5}. RAN outperforms all comparison methods at all noise levels with both symmetric and pairflip noise. It indicates that RAN can handle noisy source data well under most noise environments. Also, as mentioned earlier, RAN works well in the standard cross-scene classification task (when the noise level is 0%). The performance of the compared methods basically decreases rapidly with increasing noise levels, while RAN can yield high accuracy even at very high noise levels, especially when dealing the symmetric noise. As the noise level rises, the accuracy in pairflip noise environments decreases more rapidly than that in symmetric noise environments. Again, it demonstrates that pairflip noise is more difficult than symmetric.

6) Visualization: The visualization of the features extracted by ResNet, DAN, SPL, TCL, and RAN + MMD is shown in Fig. 6. The t-SNE embedding is used [49]. The blue and red parts denote the source and target samples, respectively. The source and target features learned by ResNet, DAN, SPL, and TCL are not aligned very well, and some samples are hard to distinguish with such confusing classification boundaries. Compared with ResNet, DAN and SPL improve the model's performance by the distribution alignment and the basic curriculum learning strategy, respectively. Their visualization results are therefore slightly better than ResNet. TCL takes into account both noise labels and cross-scene adaptation. However, blind global alignment can compromise network discriminability. As a result, the clustering performance and classification boundaries are very poor, even though the source and target domains appear to be roughly aligned. For the proposed RAN, samples of the same category are well clustered, and samples of different categories are dispersed. Furthermore, the source and target samples are aligned very well. This also demonstrates that RAN enhances the model's discriminative capability and generalizability considerably.

V. Conclusion

Cross-scene classification methods have developed rapidly in recent years and have also been applied to different areas. However, most of the existing methods assume that the source domains are noiseless, which is a very ideal setting.

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Noisy sample is very common in RS practical studies and applications, and it can severely degrade the performance of the network. In addition, the common coarse-grained global adaptation also severely limits the capability of models to cross-scene classification. It is of vital importance to solve or alleviate the above problems in practice. To achieve robust and efficient cross-scene classification under noisy environments, we propose a new plug-and-play cross-scene framework, RAN. Adopting the curriculum learning strategy and considering the class information via two weighting factors, RAN can improve the network's discriminative capability and generalizability of the network in cross-scene tasks, even when affected by noisy labels. Experiments show that RAN yields state-of-the-art results on cross-scene classification tasks under different noisy environments. Since the adaptation process and noisy sample detection are still relatively independent in the existing methods, an integration of these two steps will have the potential to enhance performance and help with the end-to-end process.

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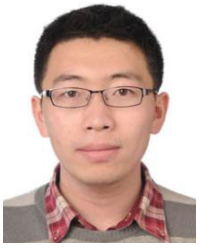
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